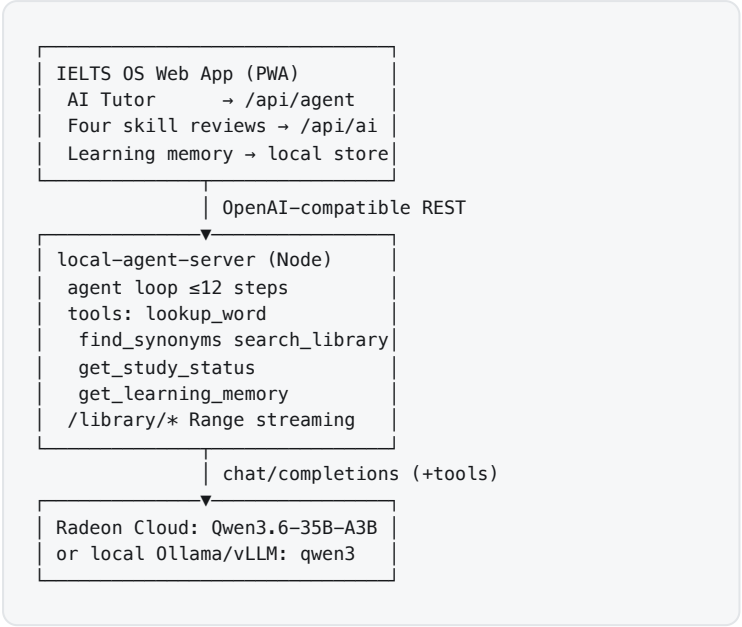
RADEON HACKATHON 2026-07 · TRACK 2

IELTS OS · Private AI Study Agent

Team **OpenSpark**
Local-first · AMD Radeon GPU · No cloud lock-in

A local-first IELTS study app with a built-in private AI agent. The tutor reads the learner's own four-skill answers, vocabulary book, material index and persistent learning memory, then turns each attempt into a review and prioritized next drill. Inference runs on AMD Radeon GPUs through Radeon Cloud or a fully local ROCm backend.

AGENT ARCHITECTURE



TRACK 2 RUBRIC COVERAGE

Capability	Implementation
Personal assistant	Long-term memory carries evidence across sessions
Knowledge-base QA	8,000-word book, 99 synonym groups, library index
Tool calling	5 local tools, visible trace in UI
Multi-step planning	≤12-step loop, parallel calls per turn
Privacy	Dual backend; keys server-side only; files streamed locally
Four-skill coach	Evidence reviews, Writing correction, Live debrief and next plan

MEASURED ON AMD RADEON GPU CLOUD

19.8 s
fresh Listening AI review used in the submitted demo

50 s
writing feedback end-to-end on the free tier

5 tools
study status, memory, vocabulary, synonyms and local materials

21
automated checks: loops, data contracts, tool calling

INFERENCE-SPEED LESSONS (FREE TIER, AUG 2026)

- Qwen3.6 hidden thinking exhausts the token budget — disabled via `chat_template_kwargs.enable_thinking=false`.
- Gateway hard-cuts at ~120 s — answers capped at 384 tokens with concise system prompts.
- Transient 5xx under load — one server-side retry plus one client retry; honest error surfacing.

github.com/Francischi-ARK/Radeon-hackathon-2026-07

Personal files and audio stay local. A local ROCm backend also keeps prompts and memory summaries on-device.